

3 VaR Methods

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2021-01-01

Historical VaR & Parametric VaR & Monte Carlo VaR These are three methods used to measure the maximum potential loss of a portfolio over a specific time horizon.

Historical VaR: Historical returns regardless of shape

Parametric VaR: Assumes retruns are normally distributed

Monte Carlo VaR: Hypothetical simulations of portfolio's future performance, assumes that future asset returns and risk factors follow stochastic processes

loading packages

```
library(quantmod)
```

```
## Loading required package: xts
```

```
## Loading required package: zoo
```

```
##
```

```
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      as.Date, as.Date.numeric
```

```
## Loading required package: TTR
```

```
## Registered S3 method overwritten by 'quantmod':
```

```
##      method      from
```

```
##      as.zoo.data.frame zoo
```

```
library(PerformanceAnalytics)
```

```
##
```

```
## Attaching package: 'PerformanceAnalytics'
```

```
## The following object is masked from 'package:graphics':
```

```
##
```

```
##      legend
```

```
library(xts)
library(ggplot2)
library(dplyr)
```

```
##
## ##### Warning from 'xts' package #####
## #
## # The dplyr lag() function breaks how base R's lag() function is supposed to #
## # work, which breaks lag(my_xts). Calls to lag(my_xts) that you type or #
## # source() into this session won't work correctly. #
## #
## # Use stats::lag() to make sure you're not using dplyr::lag(), or you can add #
## # conflictRules('dplyr', exclude = 'lag') to your .Rprofile to stop #
## # dplyr from breaking base R's lag() function. #
## #
## # Code in packages is not affected. It's protected by R's namespace mechanism #
## # Set 'options(xts.warn_dplyr_breaks_lag = FALSE)' to suppress this warning. #
## #
## #####
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:xts':
##
## first, last

## The following objects are masked from 'package:stats':
##
## filter, lag

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union
```

```
library(tinytex)
```

Retrieving Market data for selected tickers

```
symbols <- c("AAPL", "MSFT", "GOOG")

getSymbols(symbols,
           from = "2020-01-01",
           src = "yahoo")
```

```
## [1] "AAPL" "MSFT" "GOOG"
```

Extracting adjusted close prices

```
prices <- merge(Ad(AAPL),Ad(MSFT),Ad(GOOG))

colnames(prices) <- symbols

head(prices)
```

```
##              AAPL      MSFT      GOOG
## 2020-01-02 72.33386 151.8295 67.81178
## 2020-01-03 71.63063 149.9390 67.47900
## 2020-01-06 72.20143 150.3266 69.14285
## 2020-01-07 71.86185 148.9559 69.09970
## 2020-01-08 73.01783 151.3286 69.64422
## 2020-01-09 74.56879 153.2191 70.41342
```

Log Returns

```
returns <- na.omit(Return.calculate(prices,method = "log"))

head(returns)
```

```
##              AAPL      MSFT      GOOG
## 2020-01-03 -0.009769595 -0.012529957 -0.0049195848
## 2020-01-06  0.007937087  0.002581741  0.0243582011
## 2020-01-07 -0.004714386 -0.009159747 -0.0006241816
## 2020-01-08  0.015958173  0.015802909  0.0078492854
## 2020-01-09  0.021018353  0.012415214  0.0109842051
## 2020-01-10  0.002258168 -0.004637599  0.0069483269
```

Assumption: Portfolio weights AAPL: 40% MSFT: 40% GooG: 20%

```
weights <- c(0.4, 0.4, 0.2)

portfolio_returns <- Return.portfolio( returns,weights = weights)
```

Implementing Historical VaR

```
historical_var <- VaR(portfolio_returns,p = 0.95,method = "historical")

historical_var
```

```
##      portfolio_returns
## VaR      -0.02593654
```

- Interpretation: there is a 5% chance the portfolio loses more than 2.5% in one day.

Implementing Parametric VaR

Note: parametric VaR assumes the returns are normally distributed.

```
parametric_var <- VaR(portfolio_returns,p = 0.95,method = "gaussian")
parametric_var
```

```
##      portfolio_returns
## VaR      -0.02740912
```

- Interpretation: there is a 5% chance the portfolio loses more than 2.7% in one day.

Monte Carlo Simulation VaR

```
set.seed(123)

mu <- mean(portfolio_returns)
sigma <- sd(portfolio_returns)

sim_returns <- rnorm(10000,mean = mu,sd = sigma)

mc_var <- quantile(sim_returns, 0.05)

mc_var
```

```
##      5%
## -0.02736564
```

- Interpretation: There is a 5% chance the portfolio loses more than 2.7% in one day.

Visualizing return distribution

```
df <- data.frame(returns = coredata(portfolio_returns))

ggplot(df, aes(x = returns)) + geom_histogram(bins = 50) + ggtitle("Portfolio Return Distribution")

## Don't know how to automatically pick scale for object of type <xts/zoo>.
## Defaulting to continuous.

## Warning in `[<-.data.frame'('*tmp*', , x_vars, value = list(x =
## c(-0.00976959514359432, : replacement element 1 has 4815 rows to replace 1605
## rows
```

